When a Local Pricing Guard Is Not a Welfare Promise

Finite precision, continuing intervention and adaptive behavior in a bounded logit market

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Abstract. A local pricing guard can satisfy immediate profit checks while failing a customer-benefit promise over time. We study a bounded symmetric logit duopoly where permitted prices have near-best immediate profit against the last public rival price. One cap round appears protective in a coarse inherited game, but base continuous worst long-run customer benefit is about 66.272, below the unchanged relaxed 95% target of 68.513. An attained upper-price envelope characterizes fixed and mixed timing of the same physical cap. A funded calendar minimizes worst-path cap-rule frequency among causal cap-or-guard controllers protecting every activation prefix. Validated timing reduces base effective-continuous frequency by about 7.29% relative to its best fixed period; finite and continuous certificates remain separate. A complete 120-label calculation preserves all outcomes. Separately, 192 prospectively specified tabular-learning runs compare four schedules, two initializations and twelve paired seeds under two masks in one calibration. Their four paired mean customer-benefit increases are 2.39-2.95 units. A targeted 48-label follow-up holds each learned partner frozen and certifies exact achieved discounted profits and remaining response gaps. Gains are positive in 33 cases and zero in 15; profit improvements can reduce eventual-cycle customer benefit. All selected continuation paths use trained states and actions. These full-information audits do not establish equilibrium, collusive intent or discoverability. The contribution is a model-specific guarantee audit and graphical benchmark using established learning, renewal, budget and policy-response methods.

1 The problem: a local check and a long-run promise

Two shops choose prices simultaneously. Each wants its own profit. An external guard checks a proposed price against the other shop's *last completed* price. Passing that check says little by itself about what customers obtain after many executions. We ask what customer-benefit promise survives a one-time handover back to this guard, and what continuing control is needed for a stronger promise.

A cap is a real market round: it limits both executed prices, makes purchases and earns profit and customer surplus. It is not a free reset of a chart. Our human game deliberately lets a coordinator choose both shops' prices. That illustrates permitted behavior; it does not establish independent collusion or supply human experimental data.

What is added. The linked sprint starter already provides replacement, history, objective and checkpoint-branch experiments [1]. We audit an explicit external price rule: precision, attainable adverse behavior, same-cap timing, model reserve, local learning and individual responses against saved learned partners. Known control and learning methods are credited below; priority for these ingredients is not claimed.

2 A self-contained game and the scope of each claim

There are B potential buyers, two symmetric inside goods of quality v and cost c, an outside option of normalized utility zero, and taste scale mu>0. At executed physical prices p_A,p_B, let e_i=exp((v-p_i)/mu). Expected demand, private profit and customer surplus are:

```
D_i = B*e_i / (1+e_A+e_B)
R_i = (p_i-c)*D_i
C(p_A,p_B) = B*mu*log(1+e_A+e_B)
```

The base calibration is B=100, v=2, c=1 and mu=0.25. Its inherited bounded physical price domain is approximately [1.427721, 1.970186]. The inherited cap is held exactly at its saved numerical value p_cap=1.4664687420450921; it is not moved to a different continuous Nash price when timing changes. The original 15-quote reference gives cap customer benefit approximately 72.119. The unchanged relaxed 95% target F is approximately 68.513, stored as an absolute numerical threshold. Different experiments do not quietly redefine this target.

Let q be the rival's previous completed price. The guard permits every price in the bounded domain whose immediate expected profit is within epsilon of the best immediate profit available against q:

```
S(q) = {p : max_z R(z,q) - R(p,q) <= epsilon + tau}
U(q) = max S(q)
```

Epsilon=0.1 in the base mask; the learner sensitivity uses epsilon=0.05. Tau is the inherited 10^-12 numerical comparison allowance. This is a lagged one-step modeled profit-loss rule, not calibrated regret and not loss against the rival's hidden current action. Both sellers are externally constrained; they may coordinate, randomize or use arbitrary history within the legal sets when we state an *all-policy* guarantee.

Table 1. Four evidence types and their distinct scopes.

Object	What is computed	What it does not establish
Saved finite game	Exact rational comparisons on the saved binary64 payoff weights and legal masks.	Exact-real logit inequalities or grid-invariant protection.
Validated continuous model	Directed Decimal interval bounds for imported real constants under checked monotonicity and contraction conditions.	An independent transcendental backend or empirical market validity.
Learning behavior	Fixed-budget local Q-learning, paired seeds, saved-policy continuation and visit support.	Convergence, equilibrium, intent, or LLM behavior.
Frozen-partner incentives	Exact achieved discounted values and a best-response upper bound from saved histories.	Discoverability, resumed learning or every-state equilibrium.

In the graphical cap game a saved 61-quote base mesh is augmented by the exact old cap, prioritizing that quote over a nearby mesh quote. The two learner masks each have 62 choices. These are separate numerical games from the original 15-quote and refinement meshes; a continuous period is independently checked in the finite game before any finite guarantee is asserted.

Finite masks inherit the binary64-rounded epsilon+tau comparator. Continuous validation separately treats imported constants as exact reals and reports nominal/effective allowances. “Continuous limit” describes the model result, not a proof that bit-level JavaScript mask rounding vanishes.

Different questions, different outcomes

An all-policy floor concerns every allowed path. A training average concerns the actual exploratory updates. A frozen continuation stops updates and exploration and follows the saved greedy choices. These answer different questions, and the paper and game keep them visible separately.

3 The handover promise weakens with richer prices

Under the verified base continuous assumptions, U is monotone and contractive on the inherited interval. It has a unique fixed point q*. Prices on every permitted path are bounded above by iterating U, and the upper choices themselves are permitted. Since customer surplus decreases in either price, C*=C(q*,q*) is a tight worst long-run floor. This does not require every permitted price path to converge.

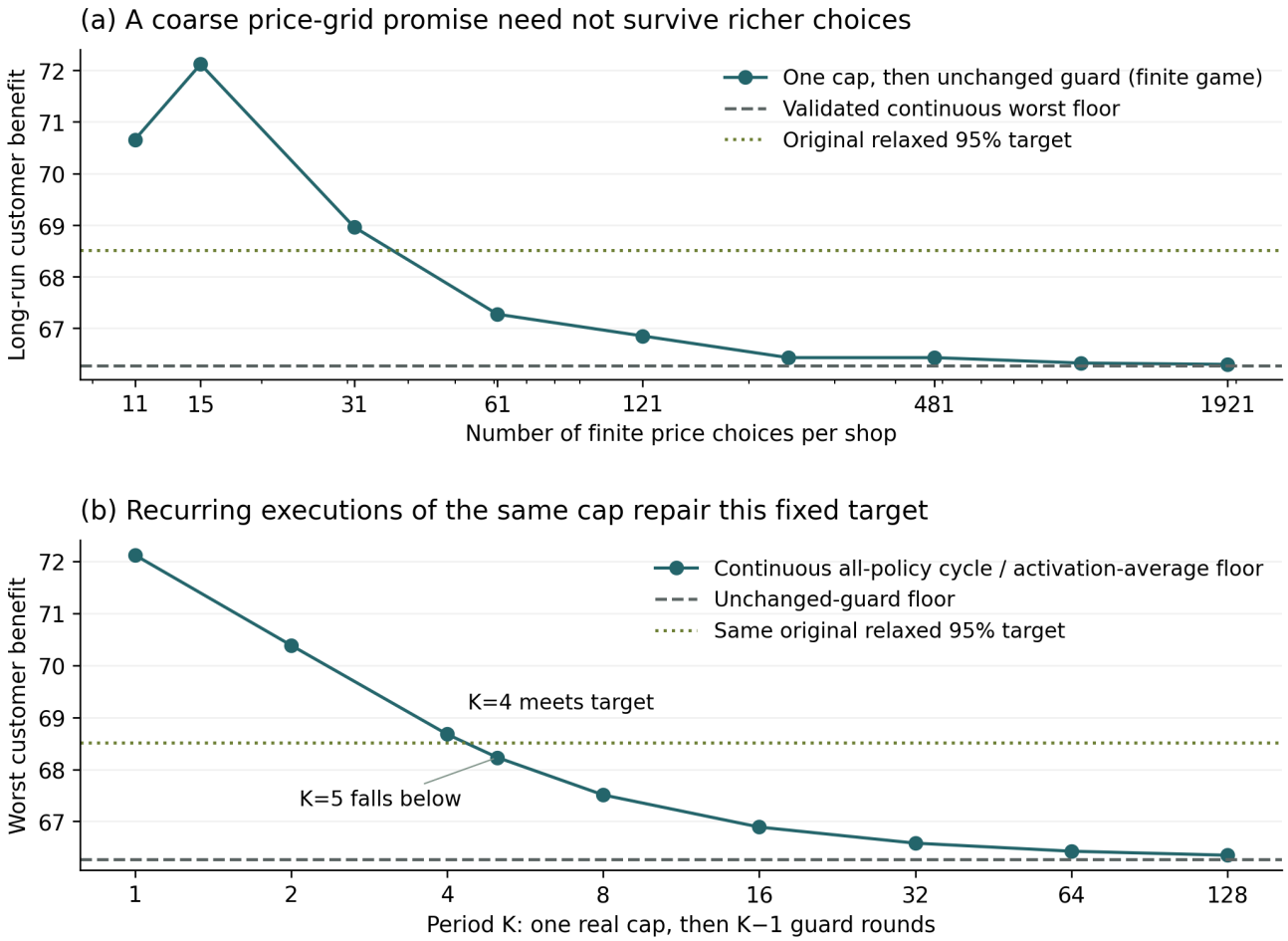


Figure 1. Base condition. (a) Saved finite one-cap floors at the predeclared mesh sizes and the validated continuous floor. Refinement is not monotone across all coarse meshes; the 11-choice case is already worse than the 15-choice case. (b) A different object: validated continuous recurring-cap floors G_K, with the same old physical cap and unchanged target. These plots show all-policy capacity, not observed learning.

The 15-choice one-cap floor is about 72.119, while the 1,921-choice diagnostic is about 66.298 and the continuous limit is 66.272. The original target is 68.513. A finite initializer changes a transient; it cannot raise the asymptotic continuous floor above C* when the unchanged guard then runs forever.

A tight repair within the fixed-period family

Keep physical price units: let a_0=p_cap and a_(j+1)=U(a_j), and put r_j=C(a_j,a_j). The actual first cap is phase 0; K-1 guard rounds follow. The worst repeated cycle is attained by both shops choosing their maximum legal quote:

```
G_K = (r_0 + ... + r_(K-1)) / K
```

The verified sequence r_j decreases, so every cap-first finite running average is at least G_K; a later individual round can still fall below it. The base target permits K=4 (G_4 about 68.676), but K=5 fails (about 68.229). Both 99% and 100% reference targets need K=1 in these audited labels. “Largest safe K” means largest *fixed integer period in this same-cap family*. Frequency 1/K counts rule applications, not economic cost, changed quotes or autonomy.

4 Can mixed gaps improve the same protection promise?

Let S_l sum the first l adverse rewards and K be the largest safe fixed period. A K-round block banks a=S_K-KF>=0; a K+1 block spends -b, where b=S_(K+1)-(K+1)F<0. Fund that entire debt before using the longer gap. Concavity then protects every activation prefix, including one ending inside a block.

```
long-block fraction p = a/(a-b)
minimum worst-path cap-rule rate = 1/(K+p)
```

The feasible maximum-price response and Jensen inequality, including the last partial gap, force this lower rate for any causal two-mode controller. Bounded funded credit attains it. This is asymptotic minimax optimality within the stated family; sliding windows and pointwise minimum cap counts are different promises. The guard remains active between caps. The methods have close prior art [9–12]; the full conditional proof is in MIXED_TIMING_THEORY.md.

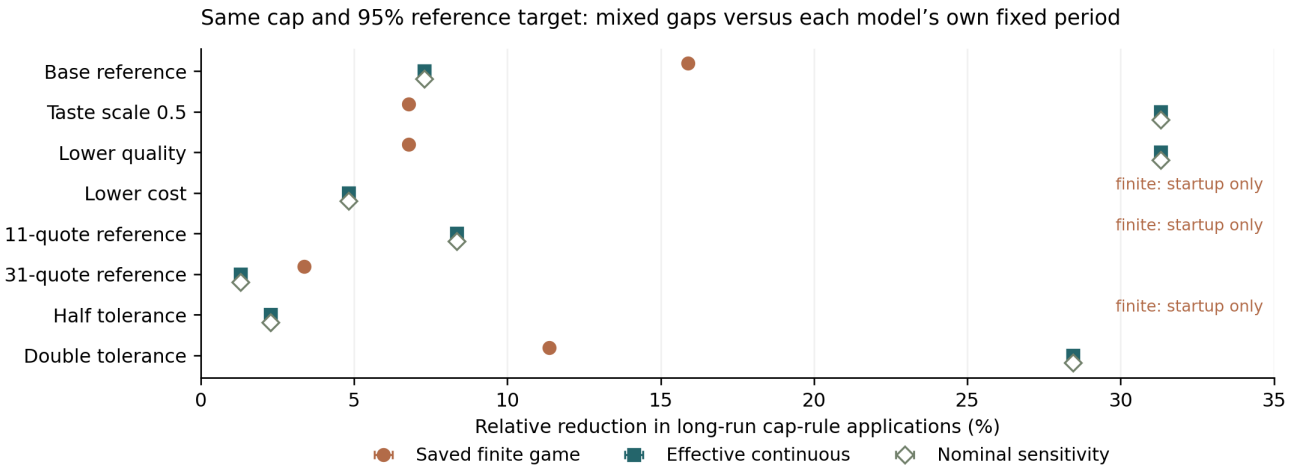


Figure 2. Prospectively selected 95% target, all eight scenarios and three model interpretations. Reductions use each record's own best fixed period. Points are constructed feasible reductions; continuous enclosures are too narrow to resolve graphically but remain nonzero. Startup-only rows have no largest fixed comparator and are labeled rather than plotted as zero savings. Reference-grid labels identify inherited conditions; finite games have 61/62 choices.

Table 2. Same physical cap and target, different numerical models. Rates are long-run cap applications per execution; decimals are approximate.

Base 95% model	Own K	Fixed rate	Constructed rate	Reduction	First long round
Saved finite	5	0.200000	0.168238223	15.8809%	6
Effective continuous	4	0.250000	0.231782998	7.2868%	13

All 120 labels remain: 40 finite give 10 startup-only cases and 30 exact adjacent optima; 80 continuous give 14 startup-only cases and 66 enclosures. All 96 adjacent savings are positive because a>0 here, not because 96 empirical trials succeeded. Every continuous rate enclosure has nonzero width (maximum below 4.19e-28). Twelve finite first-long times exceed the 256-round preview; 26 continuous times exceed 64 rounds. At the 100% target, first long gaps require millions of executions. The 192-run learner study below keeps its old fixed clocks; it supplies no behavior under this new calendar.

5 Customer protection and private incentives

The same cap and legal rule have distinct modeling and strategic boundaries. In the bounded quality-error test, base K=4 has positive reserve only to about d=0.006963. At dbar=0.01 its worst customer benefit is about 68.442, below the unchanged 68.513 target; conservative integer redesign uses K=3, or K=2 at dbar=0.05. Quality loss itself harms customers. This is one structural sensitivity, not general robustness.

Can a learned shop earn more against its frozen partner?

A targeted follow-up uses the original 62-quote mask, zero initialization, all twelve seeds, both sellers, and startup-only versus the actual saved selected95 K=4 clock. The 48 labels continue each final 2M history and NEXT phase, with no fresh cap. One seller changes its entire policy; its deterministic learned partner stays fixed. The white-box optimizer knows the complete partner, payoffs and legal supports. It does not model discovering a strategy from limited observations.

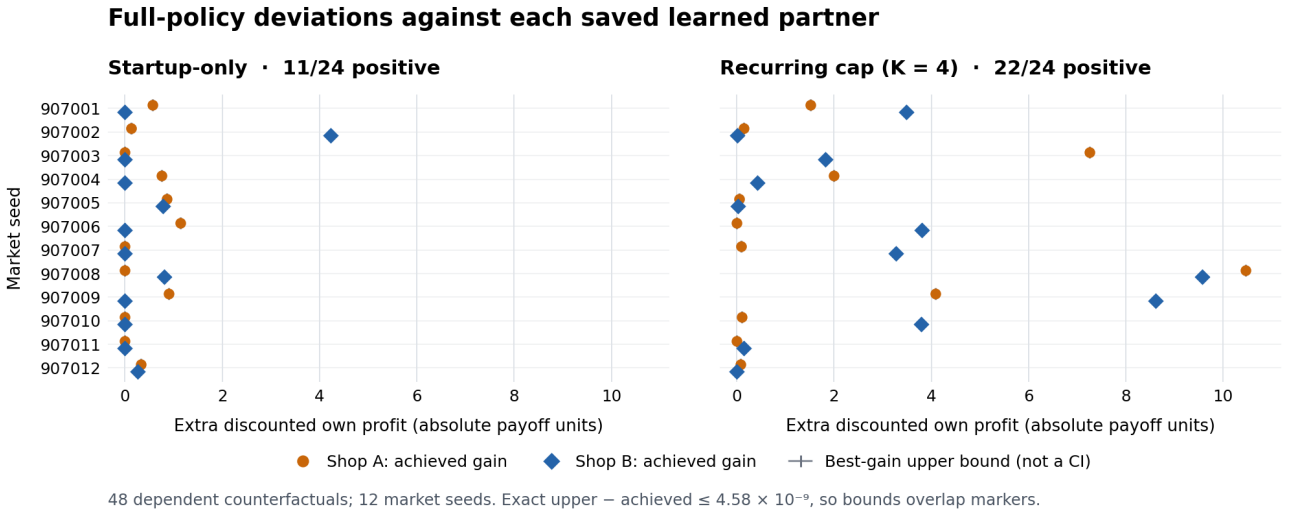


Figure 3. All 48 exact attained discounted-profit gains, both sellers and controllers. Seed identities are paired and dependent; whiskers show possible best-gain upper bounds, not confidence intervals. The maximum remaining response gap is below 4.58e-9 profit units, so the whiskers are visually indistinguishable at this scale. Zero attained gains remain visible. Gamma is mathematical 19/20; all decimals are approximate.

Table 3. Complete achieved-gain summaries in discounted profit units; exact signs, descriptive seller labels, no pooled independence.

Controller	Available	Gain + / 0 / -	Minimum	Median	Maximum
Startup, then guard	24/24	11 / 13 / 0	0.000000	0.000000	4.226564
Recurring K=4	24/24	22 / 2 / 0	0.000000	0.971166	10.459687

All 48 responses meet the prospective 1e-6 absolute gap target. Largest relative attained gains are about 0.896% of baseline discounted profit for startup and 2.355% for K=4: useful but modest. All baseline/response paths have exactly zero discounted exposure to originally unvisited states or never-tried chosen actions for either firm. This removes that particular explanation of these attained gains, not every off-path or learning concern.

Customer benefit and private gain are separate. Response-cycle customer benefit falls in 16/24 K=4 cases, rises in 6 and is unchanged in 2; startup gives 11 rises and 13 unchanged. These are eventual-cycle means, not discounted welfare or finite-prefix guarantees. A profitable allowed response does not invalidate the all-policy welfare floor or prove collusive intent. The earlier 96-label fixed-script audit remains in the evidence; its special cap-reset block simplification does not apply to learned history-dependent partners [4].

6 What independently adapting local learners do

The response audit above uses learners from the following earlier training study. To observe independent adaptation under specified rules, we froze its design before training on either game: two saved 62-price masks in the same economic calibration, four schedules, two Q initializations and twelve fresh paired whole-market seeds. All 192 runs were kept, with no outcome-driven horizon or seed changes.

Table 4. Prospectively frozen adaptive-learning design; 192 retained runs.

Factor	Predeclared values
Guard masks	Original epsilon 0.1; half tolerance epsilon 0.05. Same economic calibration.
Schedules	One startup cap then guard forever; cap every round; inherited 95%-target rhythm (K=4 or 10); K=16. All begin with an actual cap.
Private scores	Zero; optimistic value 20 times that firm's largest own immediate profit.
Seeds and budget	907001–907012, paired actor streams across conditions; 2,000,000 simultaneous rounds per run.
Learning parameters	Alpha 0.1, gamma 19/20 represented as binary64; exploration declines exponentially from 1 to 0.01, without cap/checkpoint reset.
Saved endpoints	1M secondary and 2M primary checkpoints; exact last 100k training windows; complete frozen-policy graphs; 128-step terminal previews.

Each firm stores a private Q table on (cap/guard phase, own previous price, rival previous price). Both choose before either current action is revealed. Each observes only its own deterministic expected profit for updating; no teacher, shared reward or communication scripts the choices. Current greedy selection respects the current legal mask; the bootstrap maximum uses the next state and its actual next-clock mask. Exact ties select the smallest original quote index.

```
Q(s,a) <- Q(s,a) + 0.1*[R_own + 0.95*max_(legal a_next) Q(s_next,a_next) - Q(s,a)]
```

A cap is continuing, not terminal, and the state still retains price history at cap rounds. Thus a learned cap action can depend on incoming history. Counts are saved exactly, including saturation overflow records. Two private xoshiro128** streams per seed are prospectively separated by 2^64 jump segments, and paired across conditions. Ordinary binary64 arithmetic is used for learning; there is no rational-learning claim.

The primary comparison does not secretly add a new cap

For each tolerance and initialization, compare the final frozen-greedy eventual-cycle customer benefit of the selected rhythm with startup-only, using the *actual saved terminal history and phase*. Startup-only remains in guard forever. All twelve paired differences remain in each of four strata. A new high-history, phase-0 cap start is a separately labeled counterfactual, never the primary endpoint.

The prospectively saved 5,120 by 12 whole-seed resample matrix gives approximate marginal percentile 95% seed-variation intervals, using type-7 interpolation. These are not simultaneous intervals, market uncertainty, convergence evidence or a causal effect of frequency alone. Increasing K also increases phase-dependent Q rows; equal rounds do not mean equal state/action coverage. There is no unrestricted learner-capability arm, so this design cannot show that the guard prevented learned collusion.

7 All four planned learner comparisons

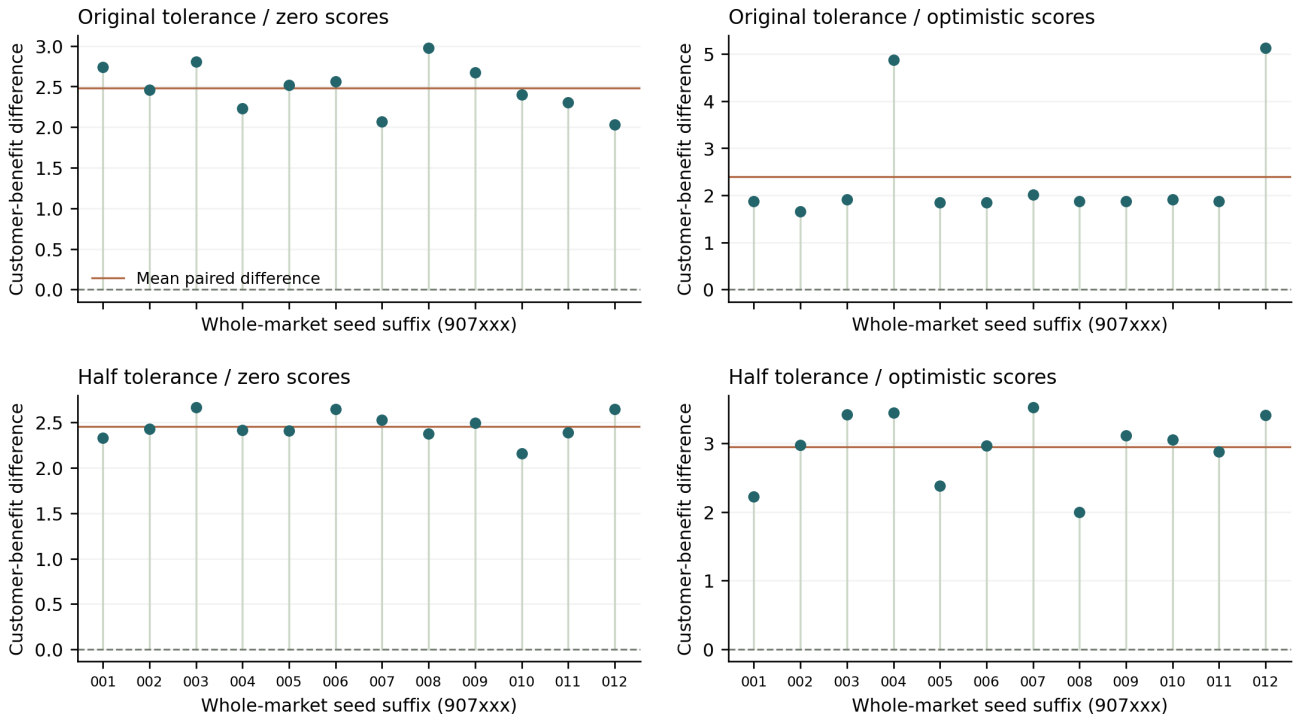


Figure 4. Every final 2M paired frozen-cycle customer-benefit difference: inherited selected95 rhythm minus one startup cap. Twelve whole-market seeds per panel; zero and reversed outcomes are retained. The horizontal line is the mean, not a protection floor. Tolerances and initializations remain separate; no seed, firm, state or checkpoint is counted as an additional independent economy.

Table 5. Final actual-terminal frozen-cycle differences, selected95 minus startup-only; all twelve paired seeds in each stratum.

Tolerance	Scores	Mean difference	All-seed range	Marginal 95% interval
Original	zero	+2.480	+2.031 to +2.978	+2.317 to +2.638
Original	optimistic	+2.391	+1.663 to +5.135	+1.850 to +3.160
Half	zero	+2.457	+2.158 to +2.664	+2.377 to +2.536
Half	optimistic	+2.946	+1.998 to +3.517	+2.657 to +3.197

Across the 48 planned paired rows, 48 differences are positive, 0 exactly zero and 0 negative. These four-stratum totals are descriptive bookkeeping, not a pooled effect estimate. Exact signs come from saved-weight fractions; intervals and printed decimals are approximate. Full seed rows remain in the evidence file.

The actual last 100k exploratory windows also favor the selected rhythm in 48/48 pairs. Four secondary mean gaps, in the paired-results table order, are +2.321, +1.979, +2.446, +3.018. These are observed training-window averages, separate from frozen cycles and the primary interval calculation.

At the final checkpoint, 0/192 actual cycles contain an unvisited state and 0/192 a never-tried greedy action. Yet 152/192 change cycle customer benefit between 1M and 2M, and 21/192 differ under fresh-cap versus actual-terminal starts. These secondary diagnostics are not independent samples or convergence evidence. The display retains exploratory windows, both profits/prices, mask geometry, visit support and separate model floors.

All 192 runs were retained: 384 million market rounds, 768 million private updates and 12,288 discarded clone steps. Execution took 331.57 seconds; the largest after-run RSS sample (407,805,952 bytes) exceeded the 256 MB soft target. It is not a measured instantaneous peak. Full resource records remain in completion.json.

Independent Python reconstructs all 384 saved-Q greedy policy pairs, complete successor graphs and exact saved-weight cycle metrics, checks actual terminal envelopes, counts and diagnostic cloned updates, and rejects corrupted evidence. The numerical core was tested before training; its full schema adapter was completed and snapshotted during training before this independent referee read its first real checkpoint. The complete main training trajectory was not independently replayed; sampled updates and cloned probes were checked. No new learner economic outcomes were interpreted before the complete learner audit passed.

8 What this adds, and what remains open

The strongest result is about *the scope of a protection promise*. A coarse finite game can give a reassuring one-cap result while a richer price interface admits worse allowed continuations. The continuous upper envelope explains the limit and gives an attainable worst cycle when the same physical cap is repeated. Calibration reserve and unilateral incentives identify different boundaries: model error can consume the old absolute target, while a worst permitted script need not be individually attractive.

The learner study adds behavior under a precisely declared controller, without replacing the all-policy proof by a favorable simulation. Its outcomes remain conditional on two masks, one calibration, chosen hyperparameters, finite budget and private-score initializations. Neither high prices nor a quiet frozen cycle establishes collusion, punishment, intent or equilibrium. The new learned-partner audit establishes modest unilateral opportunities against fixed saved policies, not mutual adaptation or punishment. Further work needs an unrestricted capability reference, matched clock capacity, learning sensitivity and tests with continued adaptation.

Relation to prior work and the sprint starter

Calvano et al. study profit-seeking learning and algorithmic pricing [2]. Eschenbaum et al. examine robustness, replacement and continued adaptation [3], and the sprint starter already branches histories, models and objectives [1]. Alshiekh et al. define action restriction and shielding, including reward credit and the need for persistent protection [5]. Brero et al. design ongoing platform interventions for consumer benefit [6]. Hartline et al. study a different, calibrated-regret regulation framework [7]. Periodic resetting and renewal-ratio control are established [8,9]. Lopes de Moraes and De Losso directly audit learned pricing incentives [4]. None should be repackaged here as our invention.

Our limited addition is a linked guarantee-capacity case: precision, a fixed physical cap, restricted minimax timing, bounded demand reserve, adaptive observations and certified learned-partner responses keep distinct quantifiers. Frozen-policy response MDPs are already explicit in [4], Sections3.2(iv),4.5; our standard Bellman bound is not a new algorithm. Adjacent timing mixtures already occur in Sun and Cyr [10]; uniform cumulative safety budgets appear in conservative bandits [11,12]. Our pricing-envelope reduction and validated comparison apply these ideas. The mixed result covers the same two actuators, not every intervention, cap strength or economic cost. No broad novelty or empirical-market superiority is established.

An accessible game, with evidence behind each view

Start at http://127.0.0.1:3003/. The two-shop challenges let a human choose both permitted quotes, see actual cap executions and separate earned customer benefit from a guaranteed buffer. The mixed-calendar page explains how earlier credit funds a longer gap. The one-shop profit challenge uses a public script and exact block score; the learner page reveals stored executions. A new one-shop continuation lets a person choose against the saved deterministic learner from its actual terminal clock; finite earned scores stay separate from audited infinite values. These are explanatory artifacts and developer-tested controls, not a human usability study.

Reproduction

Start the local viewer with `node research/demo-server.js`. Run `python research/audit-learned-opponent.py` for the new learned-partner response audit, `python research/audit-mixed-study.py` for timing, or `python research/audit-adaptive-learning.py` for saved learner artifacts. These auditors verify without rewriting closed receipts by default. README links the raw manifests, source snapshots, earlier checkers and gated derivative builders. The full archive preserves numerical evidence; the smaller viewer archive explicitly omits large checkpoints and points to the full archive hash.

Verification checks are coverage evidence, not experimental sample size. The exact finite calculations share their economic model; the continuous checkers use separate directed-interval code but the same Decimal transcendental backend. Independent AI-agent review does not substitute for external human peer review.

References

[1] Poseidon Research. Collusion hackathon starter, llm-cartel, 2026. Replacement, history, objective and checkpoint-branch code.
https://github.com/poseidon-research/collusion_hackathon/tree/main/llm-cartel

[2] E. Calvano, G. Calzolari, V. Denicolo and S. Pastorello. Artificial Intelligence, Algorithmic Pricing, and Collusion. American Economic Review 110(10), 3267-3297, 2020.
https://doi.org/10.1257/aer.20190623

[3] N. Eschenbaum, F. Mellgren and P. Zahn. Robust Algorithmic Collusion. arXiv:2201.00345,2022.
https://arxiv.org/abs/2201.00345

[4] K. Lopes de Moraes and R. De-Losso. Algorithmic Collusion Without Deterrence: Auditing the Punishment Mechanism in Q-Learning Pricing Agents. arXiv:2610.21867v2,2026.
https://arxiv.org/abs/2610.21867v2

[5] M. Alshiekh et al. Safe Reinforcement Learning via Shielding. AAAI 32(1), 2018. Sections 5.1-5.2.
https://doi.org/10.1609/aaai.v32i1.11797

[6] G. Brero, E. Mibuari, N. Lepore and D. C. Parkes. Learning to Mitigate AI Collusion on Economic Platforms. NeurIPS 35, 2022.
https://papers.neurips.cc/paper_files/paper/2022/file/f746974abd33c0015ca583a267dac1fd-Paper-Conference.pdf

[7] J. D. Hartline, C. Wang and C. Zhang. Regulation of Algorithmic Collusion, Refined: Testing Pessimistic Calibrated Regret. arXiv:2501.09740,2025.
https://arxiv.org/abs/2501.09740

[8] B. De Bruyne and F. Mori. Resetting in Stochastic Optimal Control. Physical Review Research 5, 013122, 2023.
https://doi.org/10.1103/PhysRevResearch.5.013122

[9] M. J. Neely. Fast Learning for Renewal Optimization in Online Task Scheduling. Journal of Machine Learning Research 22(279), 1-44, 2021.
https://www.jmlr.org/papers/v22/20-813.html

[10] Y. Sun and B. Cyr. Sampling for Data Freshness Optimization: Non-linear Age Functions. arXiv:1812.07241, 2019 version. Discrete two-threshold constrained sampling, Theorem 4.
https://arxiv.org/abs/1812.07241

[11] Y. Wu, R. Shariff, T. Lattimore and C. Szepesvari. Conservative Bandits. PMLR 48, 1254-1262, 2016.
https://proceedings.mlr.press/v48/wu16.html

[12] Y. Du, S. Wang and L. Huang. A One-Size-Fits-All Solution to Conservative Bandit Problems. AAAI 35(8), 7254-7261, 2021.
https://ojs.aaai.org/index.php/AAAI/article/view/16891

Event and document context

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Appendix: Limitations and Dual-Use / Ethical Considerations

Model limits. The mathematical statements require the explicitly bounded, symmetric, known logit model and checked upper-map assumptions. The saved finite results certify numerical masks and reward weights, not exact-real profit inequalities. The continuous results concern imported constants and directed intervals, not measured economic uncertainty. The quality-error test is one fixed structural perturbation, not general robustness. Frequency counts external rule applications and does not measure implementation cost or useful seller agency.

Behavioral limits. The new 192-run study has one economic calibration, two tolerance masks, twelve paired seeds and fixed learning settings. Changing phase count changes state space and coverage. Deterministic expected rewards omit sampled demand. Frozen eventual cycles may contain default unvisited scores. The 48-response follow-up freezes partners and supplies no learning-resumption, punishment-mechanism or equilibrium test. No unrestricted capability arm, convergence proof, LLM comparison or human participant study is included. White-box responses need not be discoverable from limited observations; equal training rounds do not match clock capacity or coverage. A positive fixed-script unilateral gain and a customer-benefit floor answer different questions.

Verification limits. Separate implementations reconstruct saved finite policy/graph/incentive calculations; they can share a conceptual or modeling error. Continuous primary and independent code share a Decimal transcendental backend. Learner source, checkpoints, counts and cloned probes were audited, but the complete main learning trajectory was not independently replayed; sampled updates and cloned probes were checked. Sampled after-run RSS exceeded the prospective soft memory target and is not a measured instantaneous peak. All declared runs and resource overruns remain visible.

Dual use and ethics. The toy model can illustrate adverse price coordination, so outputs should be framed as tests of safeguards rather than operational coordination advice for real firms. No real seller communication, customer data, participant recruitment, private economic data or autonomous commercial trading is involved. Human controls operate locally without claiming experimental observations. No external publication, submission, paid inference or deployment was performed in this stage.

LLM usage and author contributions

The project builds on the user's preexisting browser game and market scaffold, and on published sprint resources. Those are prior work. The guarantee-capacity analyses, new controller demonstrations, 192-run learner study and 48-case frozen-partner audit described here were developed in the current sprint work; frozen protocols, source snapshots and timestamps identify that new work. The report does not claim the inherited game or published starter as a new contribution.

Mohammadpayam Teabi supplied the project, direction and author identity. Codex and collaborating AI agents substantially performed problem formulation, code implementation, mathematical derivation, experiment orchestration, independent AI-agent checks, analysis, visualization and drafting. The report is substantially AI-written; independent AI agents are not independent human peer reviewers. No human validation of the final science is implied.

The new guarantee-capacity analyses, 192-run tabular experiment and frozen-partner response audit use no experimental language-model seller, paid model service, account-backed game player or remote inference. Codex development consumes coding-assistant allowance. Historical account-backed pilot material exists elsewhere in the earlier project and is not evidence for the claims in this report; the report does not claim that the entire project history had zero model usage.

Compact provenance pointers

Learner frozen manifest: research/adaptive-learning-results/manifest.json
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Learner independent receipt: research/adaptive-learning-results/independent-audit.json
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Appendix: Mixed-timing certificate details

The controller announces either the same full cap or the same full guard before current simultaneous quotes. The first execution is an actual cap. Cap A and each guard maximum U(q) are attainable; U(A) >= A, U is nondecreasing, and customer benefit decreases in each price. The cap-start orbit A,U(A),... gives an attained nonincreasing adverse reward list r_j tending to C*. Finite C* means this cap-start orbit, not arbitrary old guard cycles. The guarantee requires cumulative customer benefit at least F*T for every integer activation horizon T, every allowed seller policy and every realized controller random path.

For C*<F<=r_0, interpolate S_l linearly between integers. It is concave. If K is the largest integer with S_K>=KF and K+1 strictly fails, its last nonnegative crossing is L*=K+a/(a-b). Against the feasible maximum-price path, m executed caps partition any horizon T into m positive integer gaps, including the last unfinished gap. Jensen gives F*T<=sum_i S_(l_i)<=m*S(T/m), hence m/T>=1/L*. This supplies the lower bound for the worst-path limsup objective, even for history-dependent or randomized controllers. It does not impose that rate on more cooperative price paths.

For sufficiency, initialize virtual credit B=0 and choose K+1 only if B>=-b; otherwise choose K. Update credit at block completion by b or a. The invariant 0<=B<a-b implies the long-block fraction tends to a/(a-b). Each block-prefix excess S_j-jF is concave and at least its smaller endpoint, so funding the negative block endpoint protects every interior activation prefix. A partial final block cannot change the limiting rate. Credit represents a certified reward surplus, not money or actual seller cooperation. With a=0 use K only. For F<=C*, one startup cap then guard gives zero limiting rate; F>r_0 is infeasible for this cap-first family. An initial reward plateau handles F=r_0.

Finite timing uses exact rationals of saved binary64 rewards and thresholds. Continuous credits have certified bounds aL<=a<=aU, bL<=b<=bU, with aL>=0 and bU<0. Since p increases in both credits, the rate bounds are [1/(K+p(aU,bU)), 1/(K+p(aL,bL))]. Exact rational lower credits aL,bL implement the upper endpoint. This conservative feasible rate is not the unknown exact-real optimum; decimal display bounds are a separate outward rounding. All 66 interior continuous enclosures here have nonzero width.

The frozen matrix has 24 model records and 120 labels, with MAX_K=65,536 and no query at 65,537. The accepted new audit rechecks 179 sums, 2,048 stored scalar-root certificates, all finite economics/masks and exact clocks/credit/previews, with 2,452,565 checks and 14 rejected scientific corruptions. It reuses hash-bound accepted global monotonicity/contraction and fixed-point certificates rather than re-proving them; both interval implementations share Decimal transcendentals. This is independent code validation, not a formal proof assistant or external human review.

Primary computation plus final identity checks took 8.429 seconds. Largest after-model working-set sample was 110,977,024 bytes, not a continuous peak. Primary output before completion was 65,845,452 bytes, exceeding the 50,000,000-byte soft target; no labels were dropped or rerun. Derived displays and packaging are additional outputs. The first longer gap can be millions of executions away, so finite replay cap counts are not limiting-rate estimates. No new training or experimental model calls occurred.

Timing frozen manifest: research/mixed-timing-results/manifest.json
9687dd8c0664618c2cc2f3915c8e7e8ece1f4ba5dbe4873a50d2acb52f6fbc2c

Timing authoritative receipt: research/mixed-timing-results/independent-audit.json
a5fd5564d5ba6c0b0ca7d1417436a43551669a019b1cf60a7ef7b2025f65c384

Timing complete derived rows: research/mixed-timing-results/summary.json
ede197ffbe70bf5a89fcbaccc1c84b4b9b1c6715f0695ff368b9d6efa96cc4e3

Appendix: Incentives against the actual saved learned partner

This targeted follow-up selected one original-tolerance game, zero initialization, final 2M checkpoints, all 12 old seeds, both sellers and two original controllers: 48 retained labels. The stratum was chosen after the previous welfare results, before new response outcomes. It is not whole-project preregistration. The full primary and independent file/schema adapters, artificial tests, analysis contract and source snapshots were frozen before actual response computation; pre-freeze administrative checks repaired recognized legacy hash-map formats. No scientific case was rerun or omitted.

State s=(phase, previousA, previousB) retains incoming history at cap rounds. Reconstruct both original masked greedy policies from every saved binary64 Q cell, choosing the smallest original index on ties. Seller B swaps own/rival observations once; physical profitB remains in world orientation. At phase0 the cap replaces the guard; other phases use the saved rival-indexed legal support. Startup guard is absorbing. The saved selected95 clock is K=4, not the later finite K=5 or mixed calendar. Start at the actual terminal history and NEXT phase. Choices are simultaneous; the fixed rival cannot observe the focal current quote.

Numerical synchronous value iteration starts at zero, stops at completed difference 1e-10 or 600 sweeps, and proposes a binary64 vector. An isolated-process supervisor imposes 300 seconds per whole case. Final values are quantized exactly to 2^-32 with ties-to-even; exact one-step argmax selects the proposed policy over every legal action. These proposal settings are not an optimality proof. The final f64 vector is retained and its quantization/argmax checked independently; all optimizer sweeps are not independently replayed.

Lift saved binary64 rewards r=R/L and candidate values v=Z/L to a common dyadic denominator. For gamma=19/20 compute the SIGNED maximum M=max_(s,a)[20R(s,a)+19Z(s_next)-20Z(s)] over every legal edge, without clamping M to zero. Then U(s)=(Z(s)+M)/L is a Bellman supersolution. Telescoping its one-step inequality bounds every legal history-dependent or randomized focal response against this fixed rival. This is standard discounted-MDP machinery, not a new algorithm or a guarantee against an adapting partner.

For each original/proposed deterministic policy, save the full transient h and first repeated cycle c, h+c<=N. Exact value is sum_(t<h) gamma^t*r_t + gamma^h*sum_(j<c) gamma^j*r_(h+j)/(1-gamma^c). Integer polynomials evaluate the complete path, without horizon truncation. Raw gain G=V_response-V_base is kept separately from max(0,G)<=V_star-V_base<=U-V_base. U-V_response is the remaining optimality-gap bound; a zero achieved gain is not exact equilibrium. Mathematical 19/20 differs from the old training binary64 discount.

The independent audit accepted all 48 records, 24 checkpoint reconstructions, 3,660,480 compact Q cells, 553,536 world states and 3,660,480 legal state/action edges, with 19 rejected corruption controls. It checks paths, every-state response argmax/residual, count overflows, proposal quantization, execution/receipt chronology and exact seed summaries. Old economic tables are reused through accepted hashes, not re-enumerated. Full original training and optimizer trajectories remain unreplayed.

Normalized discounted unsupported exposure is (1-gamma)*sum gamma^t*I_t, using exact original state/action counts along the full transient and repeating cycle. All 48 baseline and response paths have zero exposure for each firm, both to unvisited states and never-tried chosen actions. This describes the actual audited continuations; it says nothing about every legal off-path state or retraining. Secondary customer benefit is instead an arithmetic eventual-cycle mean, excluding transient rewards, and never chooses a response or breaks an optimization tie.

All 48 executions succeeded and met the 1e-6 absolute gap target; none was loose, unavailable or over budget. Primary time was 79.727s, approximately paired sampled child/supervisor RSS 105,566,208 bytes, and primary output 19,979,290 bytes before completion. No prospective soft target was exceeded. RSS samples are neither simultaneous unique physical memory nor continuous peaks. Execution success and scientific availability are orthogonal fields; complete certificates received before a later failure would remain informative without relabeling the failed execution. No new training or model/API seller call occurred.

The 24 seller labels per controller are dependent within 12 seed identities. Exact two-seller seed means and all 12 selected-minus-startup paired differences remain in evidence, with 9 positive and 3 negative paired differences. There is no new CI or p-value. Controllers differ in learned policies, clocks, experiences and phase capacities, so this is not a causal cap-frequency effect. Full-information profitable responses do not establish discoverability, mutual adaptation, intent, subgame equilibrium or general market behavior.

Response manifest: research/learned-opponent-results/manifest.json
e1843c954dde7cc3b9d349d6664c0431fcf97839393517f3967004671bddd8d5

Authoritative response receipt: research/learned-opponent-results/independent-audit.json
9ad46a5f69e049d0c2c02ec4a6f71376722ba4c176303f8cfe4418f49849695a

Complete response summary: research/learned-opponent-results/summary.json
cab96d47d227771b5ff56476eac762cde8dcd2b1f510ca1090f75a26c496be70